

CHATTANOOGA STATE

COMMUNITY COLLEGE

CISP 1010 Computer Science1

Assignment 2

Objective: The objective of this assignment is to create an ArrayList of books in Java that can be used to add, remove, and update the elements in the list. By completing this assignment, students will gain experience working with loops, decision structures, ArrayLists, user input, and Boolean variables.

Instructions:

1. **Create a `Book` class** with the following attributes and methods:
 - **Attributes:**
 - `title` (String): the title of the book.
 - `author` (String): the author of the book.
 - `pages` (int): the number of pages in the book.
 - `rating` (double): the book's rating (out of 5).
 - **Methods:**
 - Constructors for initializing a `Book` object.
 - Getters and setters for each attribute.
 - `toString` method to provide a formatted output of the book details.
2. **Create a `BookUtility` class** with two static methods for the following tasks:
 - **Calculate the average rating** for the books in the ArrayList.
 - **Find the book with the highest number of pages** from the provided ArrayList.
3. **In the main method, using the `Book` and `BookUtility` classes, do the following:**
 - Create an ArrayList of books.
 - Provide the following menu options for the user:
 1. Add a book.
 2. Update a book's details.
 3. Remove a book.
 4. List all books.
 5. Calculate the average rating of the books.
 6. Display the details of the book with the highest number of pages.
 7. Exit.

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- The program should perform the appropriate action based on the user's choice.
 - The program should stop if the user selects the "Exit" option.
4. **Use the methods from the `BookUtility` class** for options 5 and 6.
 5. **Write Javadoc comments** and generate the HTML documentation for the `Book` and `BookUtility` classes.

Submission:

Submit the following:

- Java source code files (`Book`, `BookUtility`, and the tester/main class).
- Documentation files generated by Javadoc for the `Book` and `BookUtility` classes.

Happy coding!